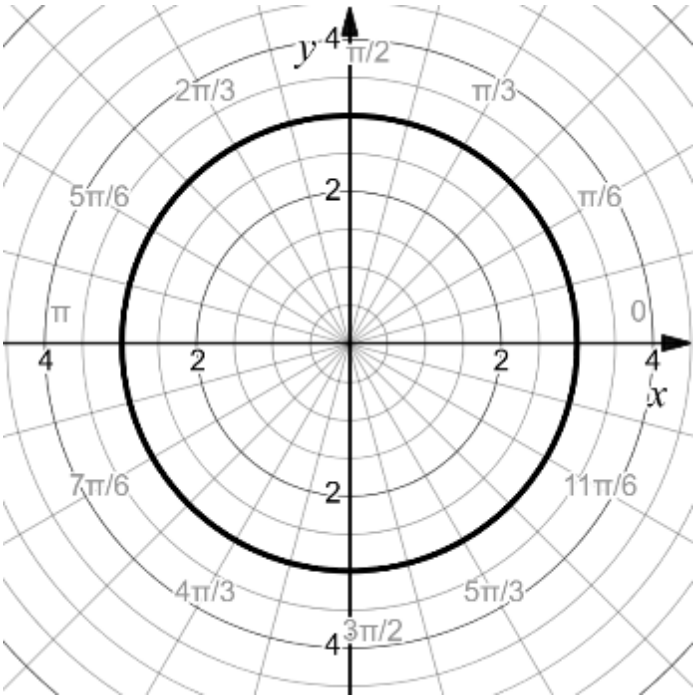
1.28 Graph Sketching - Polar Curves

Worked Solutions

1 a)



1 b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	$\sqrt{2}$	0	$-\sqrt{2}$	-1	$-\sqrt{2}$	0	$\sqrt{2}$	1

1 c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	$\frac{3\sqrt{2}}{2}$	3	$\frac{3\sqrt{2}}{2}$	0	$-\frac{3\sqrt{2}}{2}$	-3	$-\frac{3\sqrt{2}}{2}$	0

1 d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	-5	$-\frac{5\sqrt{2}}{2}$	0	$\frac{5\sqrt{2}}{2}$	5	$\frac{5\sqrt{2}}{2}$	0	$-\frac{5\sqrt{2}}{2}$	-5

1 e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	$-\sqrt{2}$	-2	$-\sqrt{2}$	0	$\sqrt{2}$	2	$\sqrt{2}$	0

2 a)

$r = 5$

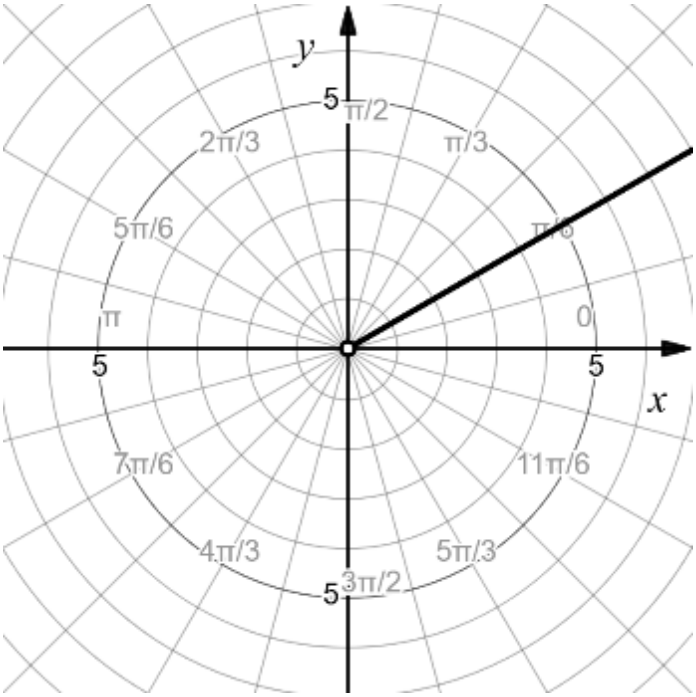
2 b)

$r = 4\cos\theta$

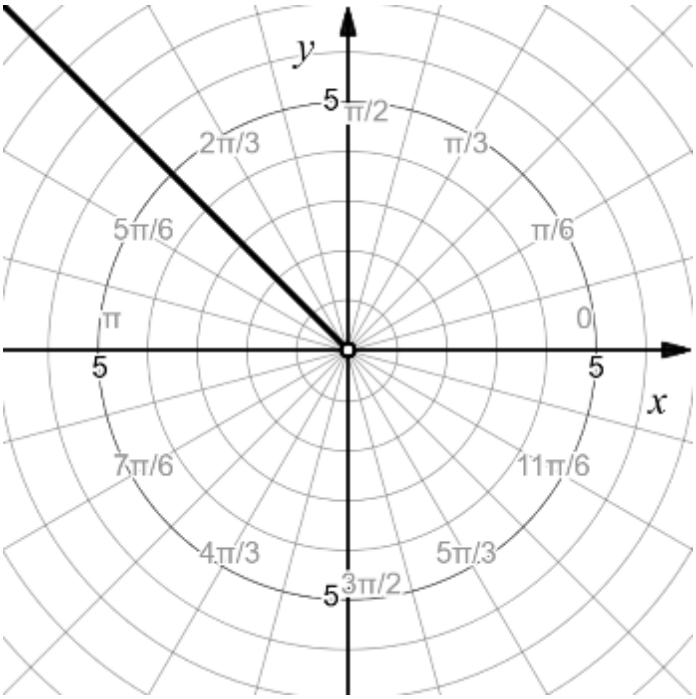
2 c)

$r = 3\sin\theta$

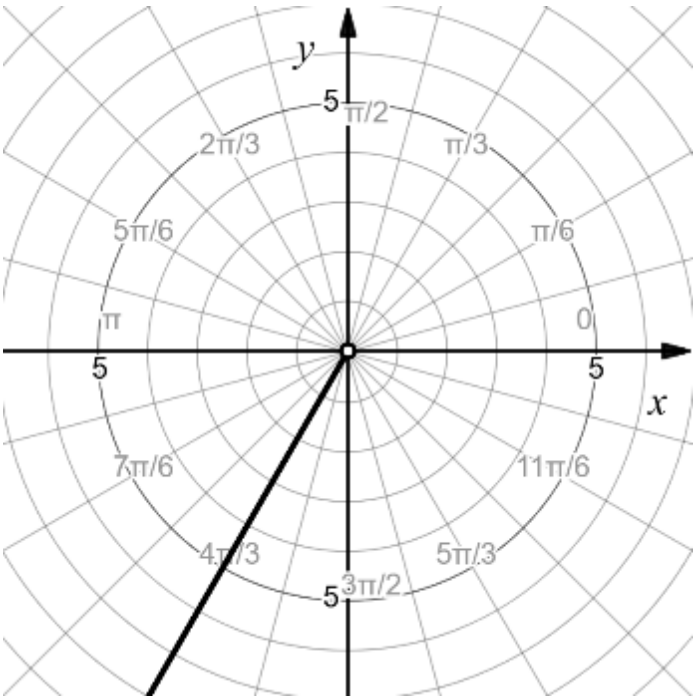
3 a)



3 b)



3 c)



4 a)

$\theta = \frac{\pi}{3}$

4 b)

$\theta = \frac{5\pi}{6}$

4 c)

$\theta = -\frac{4\pi}{3}$

5 a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	6	5.12	3	0.879	0	0.879	3	5.12	6

5 b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	2	3.41	4	3.41	2	0.586	0	0.586	2

5 c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	0.293	1	1.71	2	1.71	1	0.293	0

5 d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	4	1.17	0	1.17	4	6.83	8	6.83	4

5 e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	4	3.41	2	0.586	0	0.586	2	3.41	4

6 a)

$r = 2.5(1 + \sin\theta)$

6 b)

$r = 1.5(1 - \cos\theta)$

6 c)

$r = 2(1 + \cos\theta)$

7 a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	3	2.41	1	-0.414	-1	-0.414	1	2.41	3

$0 = 1 + 2\cos\theta$ has solutions $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$ so these are the two values between which r is negative.

7 b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	2	4.83	6	4.83	2	-0.828	-2	-0.828	2

$0 = 2 + 4\sin\theta$ has solutions $\theta = \frac{7\pi}{6}$ and $\theta = \frac{11\pi}{6}$ so these are the two values between which r is negative.

7 c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	4	3.12	1	-1.12	-2	-1.12	1	3.12	4

$0 = 1 + 3\cos\theta$ has solutions $\theta = 1.91$ and $\theta = 4.37$ so these are the two values between which r is negative.

7 d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	-1	-0.121	2	4.12	5	4.12	2	-0.12	-1

$0 = 2 - 3\cos\theta$ has solutions $\theta = 0.841$ and $\theta = 5.44$ so r is negative up to 0.841 and after 5.44

7 e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	-2.54	-4	-2.54	1	4.54	6	4.54	1

$0 = 1 - 5\sin\theta$ has solutions $\theta = 0.201$ and $\theta = 2.94$ so these are the two values between which r is negative.

8 a)

Maximum $r = 5$

Internal loop has $r = -1$

So sine with amplitude 3

$$r = 2 - 3\sin\theta$$

8 b)

Maximum $r = 4$

Internal loop has $r = -2$

So cos with amplitude 3

$$r = 1 + 3\cos\theta$$

8 c)

Maximum $r = 3$

Internal loop has $r = -1$

So cos with amplitude 2

$$r = 1 - 2\cos\theta$$

9 a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	3	2.71	2	1.29	1	1.29	2	2.71	3

9 b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	5	4.41	3	1.59	1	1.59	3	4.41	5

9 c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	4	5.41	6	5.41	4	2.59	2	2.59	4

9 d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	1.59	3	4.41	5	4.41	3	1.59	1

9 e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1.5	0.793	0.5	0.793	1.5	2.21	2.5	2.21	1.5

10 a)

Maximum $r = 3$

Minimum $r = 1$

So sin with amplitude 1

$$r = 2 + \sin\theta$$

10 b)

Maximum $r = 4$

Minimum $r = 2$

So cos with amplitude 1

$$r = 3 + \cos\theta$$

10 c)

Maximum $r = 6$

Minimum $r = 1$

So cos with amplitude 2.5

$$r = 3.5 - 2.5\cos\theta$$

11 a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	1	0	-1	0	1	0	-1	0

$0 = \sin 2\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{2}$, $\theta = \pi$, $\theta = \frac{3\pi}{2}$ and $\theta = 2\pi$ so these are the five values at which r swaps between positivity.

11 b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	0.707	-1	0.707	0	-0.707	1	-0.707	0

$0 = \sin 3\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{3}$, $\theta = \frac{2\pi}{3}$, $\theta = \pi$, $\theta = \frac{4\pi}{3}$, $\theta = \frac{5\pi}{3}$ and $\theta = 2\pi$ so these are the seven values at which r swaps between positivity.

However, the 6 'petals' overlap leaving 3 'petals'.

11 c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	0	0	0	0	0	0	0	0

$0 = \sin 4\theta$ has solutions given in the table above, these are the nine values at which r swaps between positivity.

11 d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	-0.707	1	-0.707	0	0.707	-1	0.707	0

$0 = \sin 5\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{5}$, $\theta = \frac{2\pi}{5}$, $\ldots$, $\theta = \frac{7\pi}{5}$, $\theta = 2\pi$ so these are the eleven values at which r swaps between positivity.

However, the 10 'petals' overlap leaving 5 'petals'.

11 e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	3	0	-3	0	3	0	-3	0

$0 = 3\sin 2\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{2}$, $\theta = \pi$, $\theta = \frac{3\pi}{2}$ and $\theta = 2\pi$ so these are the five values at which r swaps between positivity.

11 f)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	2.83	-4	2.83	0	-2.83	4	-2.83	0

$0 = 4\sin 3\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{3}$, $\theta = \frac{2\pi}{3}$, $\theta = \pi$, $\theta = \frac{4\pi}{3}$, $\theta = \frac{5\pi}{3}$ and $\theta = 2\pi$ so these are the seven values at which r swaps between positivity.

However, the 6 'petals' overlap leaving 3 'petals'.

11 g)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	0	0	0	0	0	0	0	0

$0 = -\sin 4\theta$ has solutions given in the table above, these are the nine values at which r swaps between positivity, starting with negative values of r

11 h)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	2.12	-3	2.12	0	-2.12	3	-2.12	0

$0 = -3\sin 5\theta$ has solutions $\theta = 0$, $\theta = \frac{\pi}{5}$, $\theta = \frac{2\pi}{5}$, ..., $\theta = \frac{7\pi}{5}$, $\theta = 2\pi$ so these are the eleven values at which r swaps between positivity.

However, the 10 'petals' overlap leaving 5 'petals'.

12 a)

k

12 b)

$-k$

12 c)

a times

12 d)

$2a$ times

13 a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	0	-1	0	1	0	-1	0	1

$0 = \cos 2\theta$ has solutions $\theta = \frac{\pi}{4}$, $\theta = \frac{3\pi}{4}$, $\theta = \frac{5\pi}{4}$ and $\theta = \frac{7\pi}{4}$ so these are the four values at which r swaps between positivity.

13 b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	-0.707	0	0.707	-1	0.707	0	-0.707	1

$0 = \cos 3\theta$ has solutions $\theta = \frac{\pi}{6}$, $\theta = \frac{3\pi}{6}$, $\theta = \frac{5\pi}{6}$, $\theta = \frac{7\pi}{6}$, $\theta = \frac{9\pi}{6}$ and $\theta = \frac{11\pi}{6}$ so these are the six values at which r swaps between positivity.

The 3 'petals' overlap so that there are both positive and negative values of r on this graph.

13 c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	-1	1	-1	1	-1	1	-1	1

$0 = \cos 4\theta$ has solutions $\theta = \frac{\pi}{8}$, $\theta = \frac{3\pi}{8}$, $\theta = \frac{5\pi}{8}$, …, $\theta = \frac{13\pi}{8}$, $\theta = \frac{15\pi}{8}$ so these are the eight values at which r swaps between positivity.

13 d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	-0.707	0	0.707	-1	0.707	0	-0.707	1

$0 = \cos 5\theta$ has solutions $\theta = \frac{\pi}{10}$, $\theta = \frac{3\pi}{10}$, $\theta = \frac{5\pi}{10}$, …, $\theta = \frac{17\pi}{10}$, $\theta = \frac{19\pi}{10}$ so these are the ten values at which r swaps between positivity.

The 5 'petals' overlap so that there are both positive and negative values of r on this graph.

13 e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	3	0	-3	0	3	0	-3	0	3

$0 = 3\cos 2\theta$ has solutions $\theta = \frac{\pi}{4}$, $\theta = \frac{3\pi}{4}$, $\theta = \frac{5\pi}{4}$ and $\theta = \frac{7\pi}{4}$ so these are the four values at which r swaps between positivity.

13 f)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	4	-2.83	0	2.83	-4	2.83	0	-2.83	4

$0 = 4\cos 3\theta$ has solutions $\theta = \frac{\pi}{6}$, $\theta = \frac{3\pi}{6}$, $\theta = \frac{5\pi}{6}$, $\theta = \frac{7\pi}{6}$, $\theta = \frac{9\pi}{6}$ and $\theta = \frac{11\pi}{6}$ so these are the six values at which r swaps between positivity.

The 3 'petals' overlap so that there are both positive and negative values of r on this graph.

13 g)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	-1	1	-1	1	-1	1	-1	1	-1

$0 = \cos 4\theta$ has solutions $\theta = \frac{\pi}{8}$, $\theta = \frac{3\pi}{8}$, $\theta = \frac{5\pi}{8}$, …, $\theta = \frac{13\pi}{8}$, $\theta = \frac{15\pi}{8}$ so these are the eight values at which r swaps between positivity.

13 h)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	-3	2.12	0	-2.12	3	-2.12	0	2.12	-3

$0 = -3\cos 5\theta$ has solutions $\theta = \frac{\pi}{10}$, $\theta = \frac{3\pi}{10}$, $\theta = \frac{5\pi}{10}$, …, $\theta = \frac{17\pi}{10}$, $\theta = \frac{19\pi}{10}$ so these are the ten values at which r swaps between positivity.

The 5 'petals' overlap so that there are both positive and negative values of r on this graph.

14 a)

k

14 b)

$-k$

14 c)

a times

14 d)

$2a$ times

15 a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	2	0	-2	0	2	0	-2	0

$0 = 2\sin 2\theta$ has solutions $\theta = \frac{\pi}{2}$, $\theta = \pi$ and $\theta = \frac{3\pi}{2}$ so these are the three values at which r swaps between positivity.

15 b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	3	0	-3	0	3	0	-3	0	3

$0 = 3\cos 2\theta$ has solutions $\theta = \frac{\pi}{4}$, $\theta = \frac{3\pi}{4}$, $\theta = \frac{5\pi}{4}$ and $\theta = \frac{7\pi}{4}$ so these are the four values at which r swaps between positivity.

15 c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	2.82	-4	2.82	0	-2.82	4	-2.82	0

$0 = 4\sin 3\theta$ has solutions $\theta = \frac{\pi}{3}$, $\theta = \frac{2\pi}{3}$, $\theta = \pi$, $\theta = \frac{4\pi}{3}$ and $\theta = \frac{5\pi}{3}$ so these are the five values at which r swaps between positivity.

However, the 6 'petals' overlap leaving 3 'petals'.

15 d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	2	-1.41	0	1.41	-2	1.41	0	-1.41	2

$0 = 2\cos 5\theta$ has solutions $\theta = \frac{\pi}{10}$, $\theta = \frac{3\pi}{10}$, $\theta = \frac{5\pi}{10}$, $\theta = \frac{7\pi}{10}$, $\theta = \frac{9\pi}{10}$, $\theta = \frac{11\pi}{10}$, $\theta = \frac{13\pi}{10}$, $\theta = \frac{15\pi}{10}$, $\theta = \frac{17\pi}{10}$ and $\theta = \frac{19\pi}{10}$ so these are the ten values at which r swaps between positivity.

However, the 10 'petals' overlap leaving 5 'petals'.

15 e)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	-1	0	1	0	-1	0	1	0

$0 = \sin 6\theta$ has solutions $\theta = \frac{\pi}{6}$, $\theta = \frac{2\pi}{6}$, $\theta = \frac{3\pi}{6}$, $\theta = \frac{4\pi}{6}$, $\theta = \frac{5\pi}{6}$, $\theta = \pi$, $\theta = \frac{7\pi}{6}$, $\theta = \frac{8\pi}{6}$, $\theta = \frac{9\pi}{6}$, $\theta = \frac{10\pi}{6}$, $\theta = \frac{11\pi}{6}$ and $\theta = 2\pi$ so these are the twelve values at which r swaps between positivity.

16 a)

Cosine with 5 petals:

$$r = 4\cos 5\theta$$

16 b)

Sine with 16 petals:

$$r = 2\sin 8\theta$$

16 c)

Sine with 4 petals:

$$r = 4\sin 2\theta$$

16 d)

Cosine with 8 petals:

$$r = 5\cos 4\theta$$

16 e)

Cosine with 4 petals:

$$r = 3\cos 2\theta$$

16 f)

Sine with 3 petals:

$$r = 5\sin 3\theta$$

17 a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	0	0.393	0.785	1.19	1.57	1.96	2.36	2.75	3.14

17 b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	-1.57	-1.18	-0.785	-0.392	0	0.392	0.785	1.18	1.57

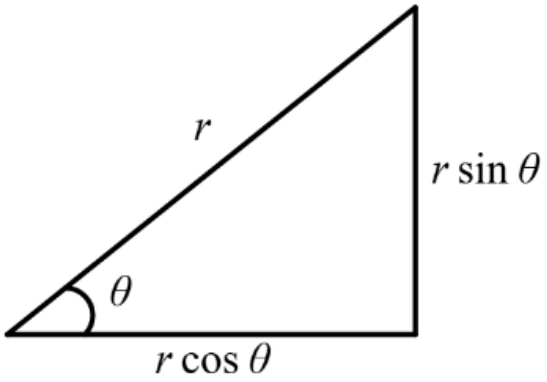
18)

The gaps between the spirals on the curve become closer together.

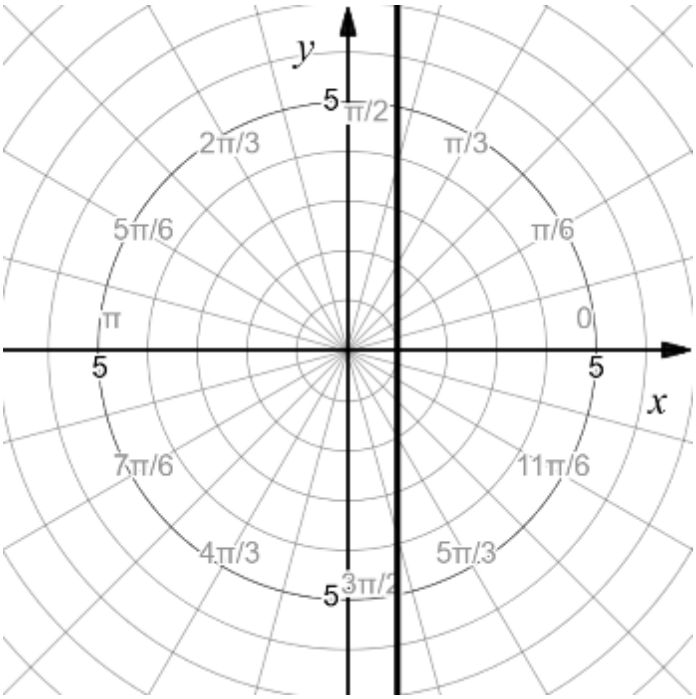
19 a)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	1	$\sqrt{2}$		$-\sqrt{2}$	-1	$-\sqrt{2}$		$\sqrt{2}$	1

By considering the right-angle triangle:



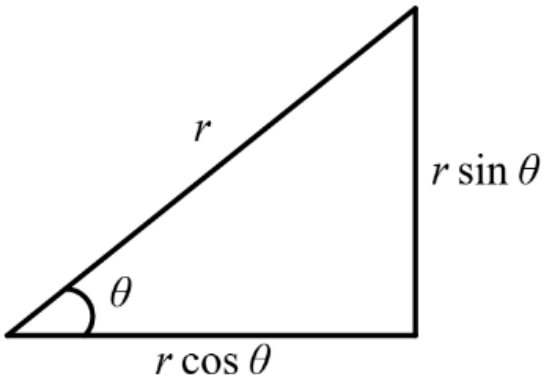
Since $r\cos\theta = 1$ the horizontal distance is always 1



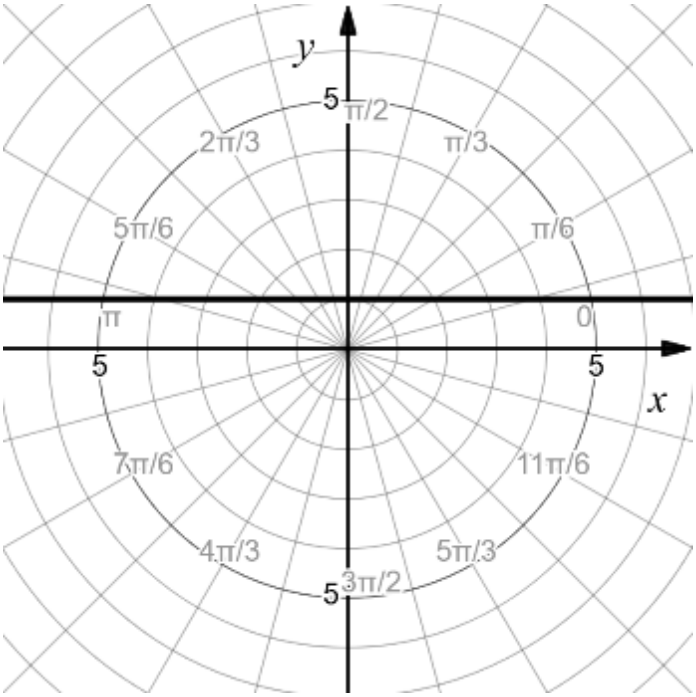
19 b)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$		$\sqrt{2}$	1	$\sqrt{2}$		$-\sqrt{2}$	-1	$-\sqrt{2}$	

By considering the right-angle triangle:



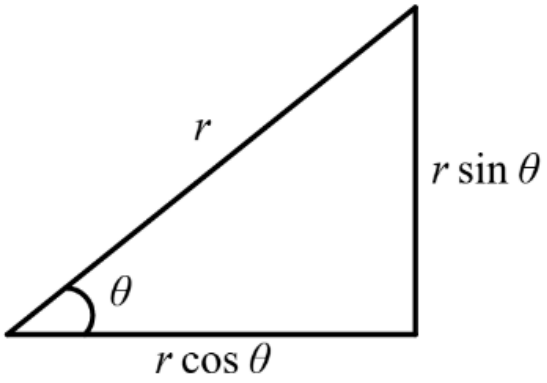
Since $r \sin\theta = 1$ the vertical distance is always 1



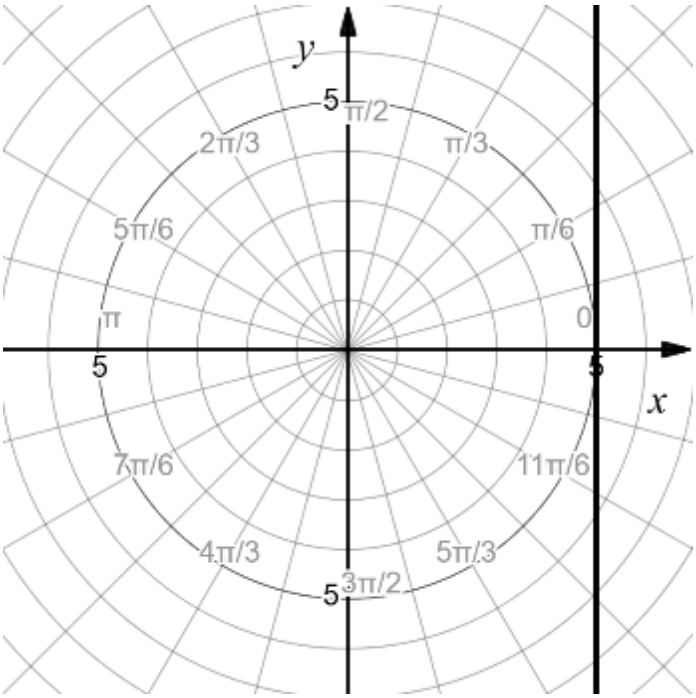
19 c)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$	5	$5\sqrt{2}$		$-5\sqrt{2}$	-5	$-5\sqrt{2}$		$5\sqrt{2}$	5

By considering the right-angle triangle:



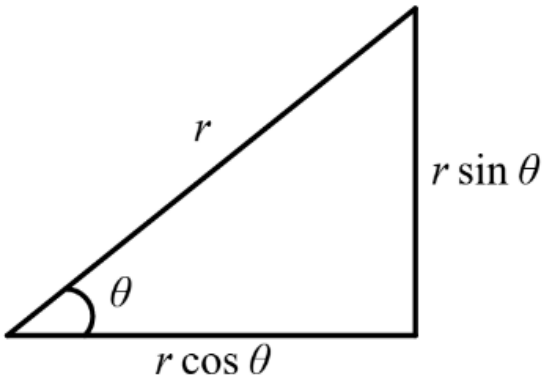
Since $r\cos\theta = 5$ the horizontal distance is always 5



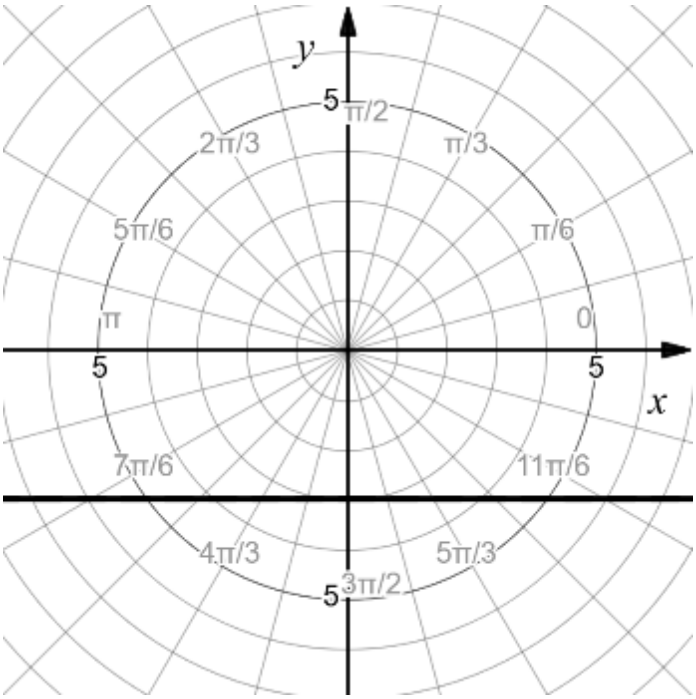
19 d)

$\boldsymbol{\theta}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\boldsymbol{r}$		$-3\sqrt{2}$	-3	$-3\sqrt{2}$		$3\sqrt{2}$	3	$3\sqrt{2}$	

By considering the right-angle triangle:



Since $r\sin\theta = -3$ the vertical distance is always -3



20 a)

$r = \frac{3}{\cos\theta}$

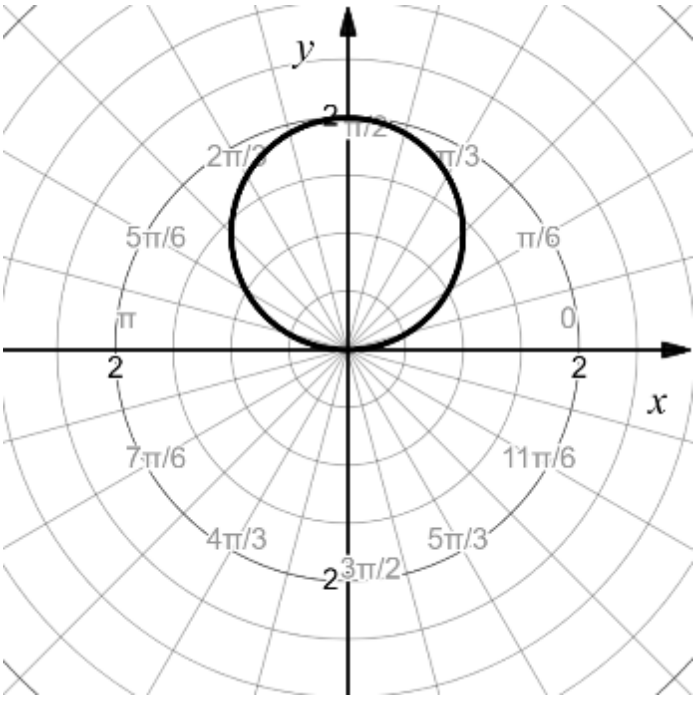
20 b)

$r = \frac{4}{\sin\theta}$

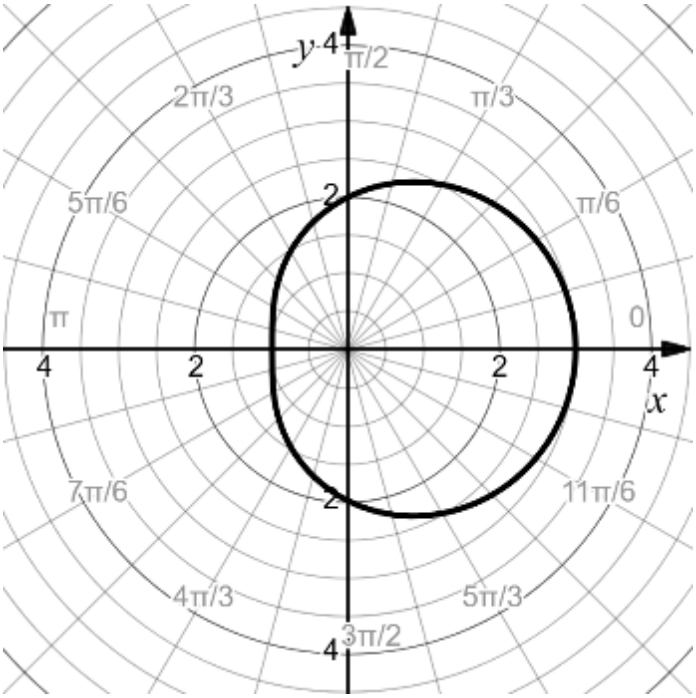
20 c)

$r = -\frac{2}{\cos\theta}$

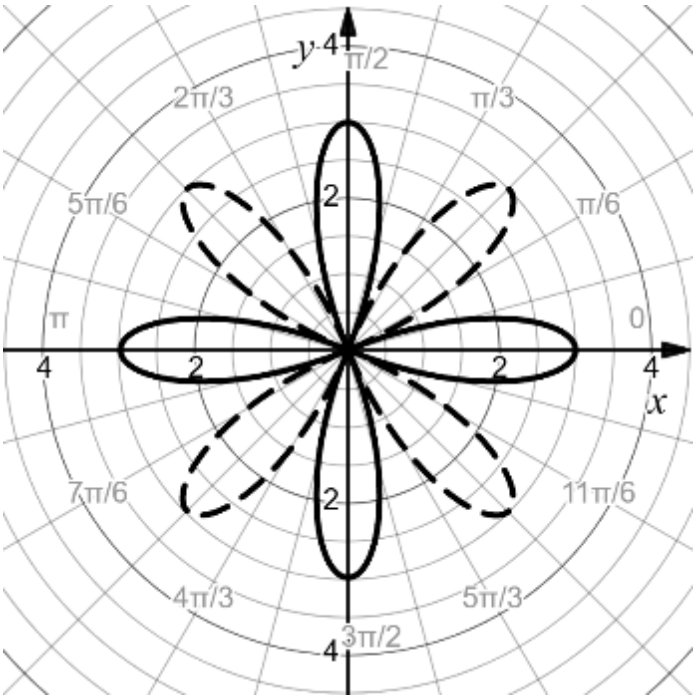
21 a)



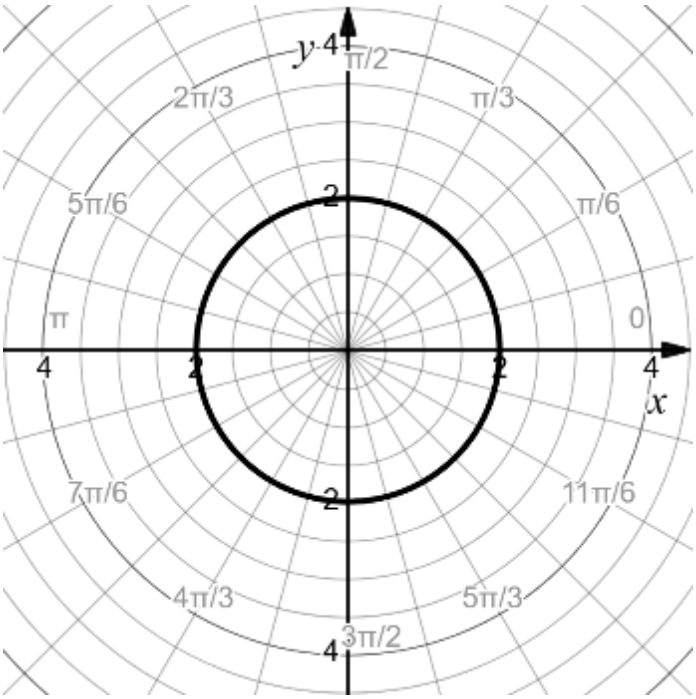
21 b)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
4
2
2
4
2
4

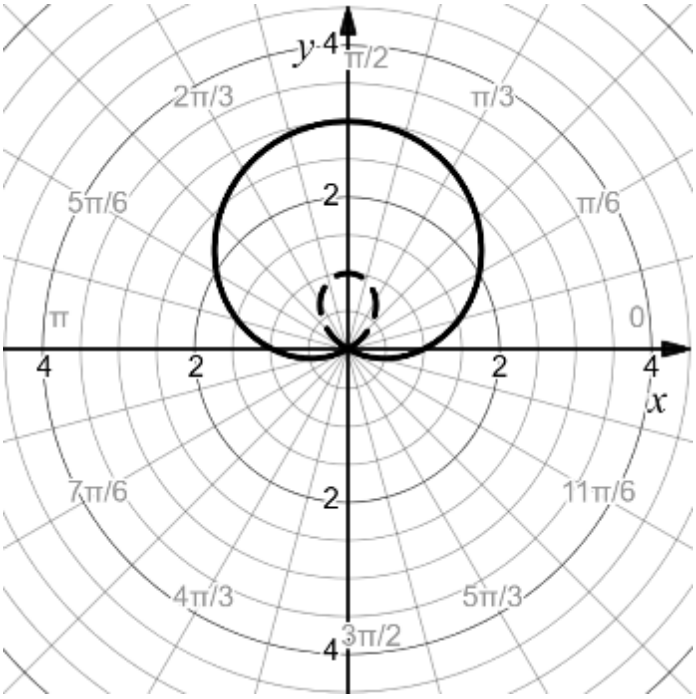
21 c)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
4
2
2
4
2
4

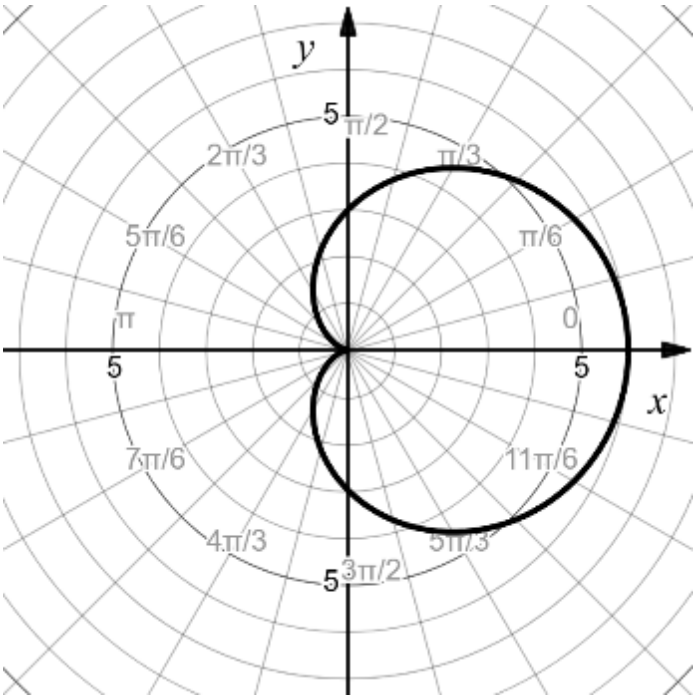
21 d)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
4
2
2
4
2
2
4

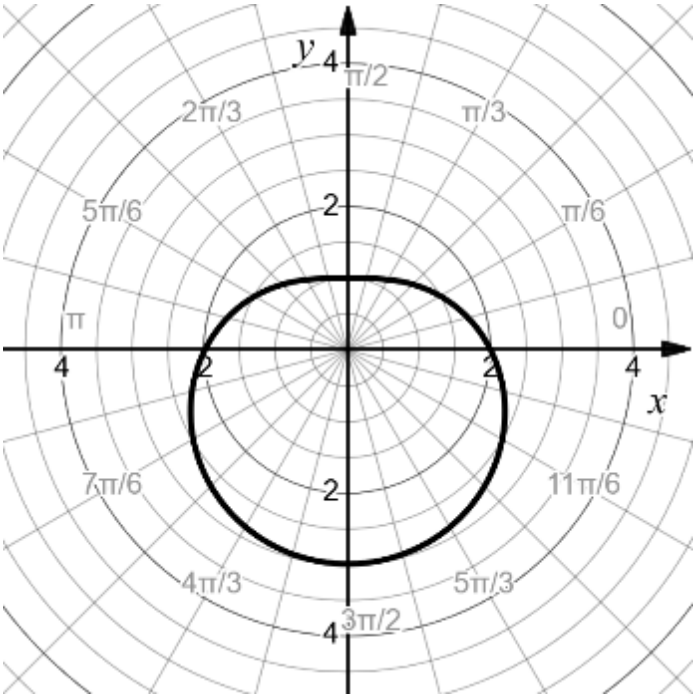
21 e)


y
x
π/2
π/3
π/6
0
2π/3
5π/6
π
7π/6
11π/6
4π/3
5π/3
3π/2
4
2
2
4
2
2
4

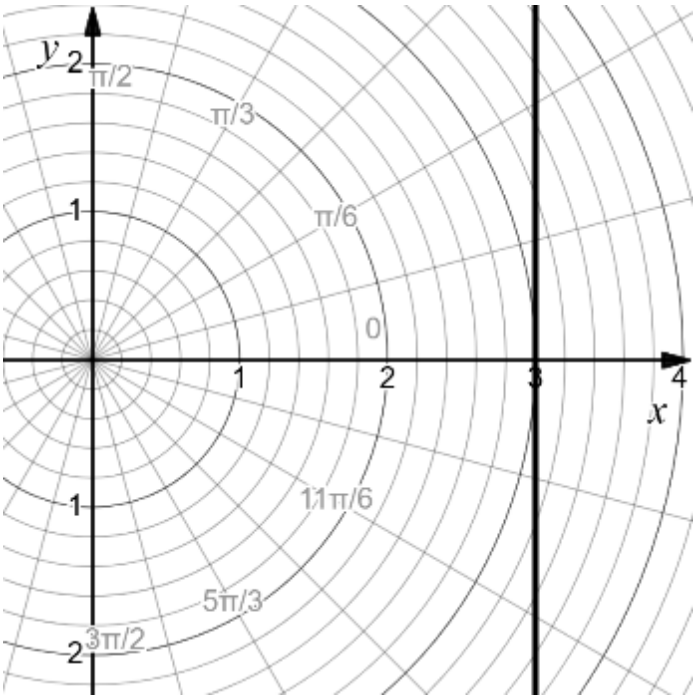
21 f)


y
x
5
π/2
2π/3
π/3
5π/6
π/6
π
0
5
5
7π/6
11π/6
4π/3
5π/3
5
3π/2

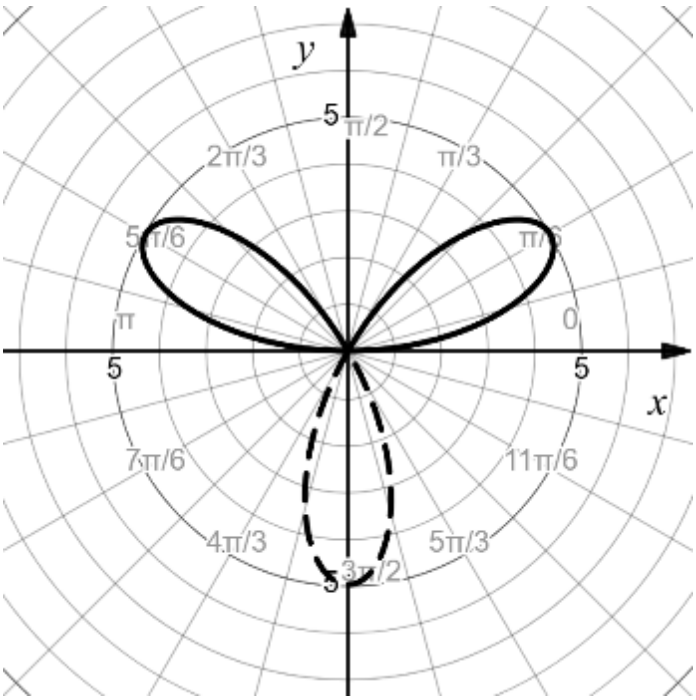
21 g)


y
x
4
π/2
2π/3
π/3
2
5π/6
π/6
π
0
4
2
2
4
7π/6
11π/6
2
4π/3
5π/3
4
3π/2

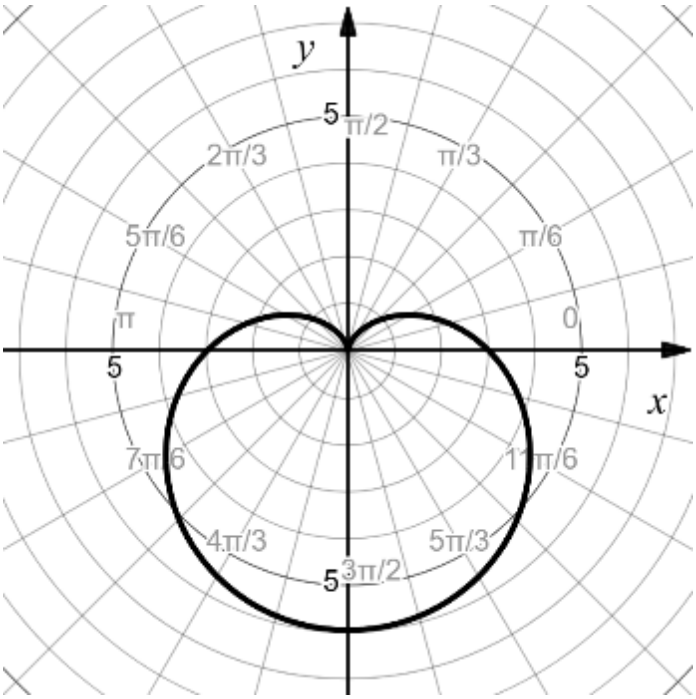
21 h)


y
x
π/2
π/3
π/6
0
11π/6
5π/3
3π/2
1
2
3
4

21 i)


y
x
5
π/2
2π/3
π/3
π
0
5
5
7π/6
11π/6
4π/3
5π/3
3π/2

21 j)



22 a)

$r = \frac{a}{\sin\theta}$ where $\theta \neq n\pi$ for $n \in \mathbb{Z}$

22 b)

$r = \frac{2a}{\cos\theta}$ where $\theta \neq \frac{n\pi}{2}$ for $n \in \mathbb{Z}$

22 c)

$r = a\cos\theta$

22 d)

$r = 2a\sin\theta$

22 e)

$r = a + a\cos\theta$ or $r = a - a\cos\theta$

Both of these pass through $(0, a)$

$$\therefore r = a + a\cos\theta \text{ or } r = a - a\cos\theta$$

22 f)

$r = \frac{a}{2} + \frac{a}{2}\sin\theta$ or $r = \frac{a}{2} - \frac{a}{2}\sin\theta$

Both of these pass through $(\frac{a}{2}, 0)$

$$\therefore r = \frac{a}{2} + \frac{a}{2}\sin\theta \text{ or } r = \frac{a}{2} - \frac{a}{2}\sin\theta$$

23 a)

Sine with 8 petals:

$$r = 4\sin 4\theta$$

23 b)

$r = 2.5(1 - \sin\theta)$

23 c)

Maximum $r = 5$

Internal loop has $r = -1$

So cosine with amplitude 3

$$r = 2 + 3\cos\theta \text{ or } r = -2 + 3\cos\theta$$

23 d)

Cosine with 7 petals:

$$r = 5\cos 7\theta$$

24)

$\sin(\theta - \frac{\pi}{6})$ for $\frac{\pi}{6}$ to $\frac{4\pi}{6}$ increases from 0 to 1

$\sin(\theta - \frac{\pi}{6})$ for $\frac{4\pi}{6}$ to $\frac{7\pi}{6}$ decreases from 1 to 0

At $\theta = \frac{2\pi}{3}$ then $r = 2$

So the shortest distance to the line is $r = 2$ which is at angle $\theta = \frac{2\pi}{3}$

A straight line where:

θ	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$	$\frac{5\pi}{6}$	$\frac{6\pi}{6}$
r	4	$\frac{4\sqrt{3}}{3}$	2	$\frac{4\sqrt{3}}{3}$	4

At $\theta = \pi$ then $r = 4$

At $\theta = \frac{\pi}{2}$ then $r = \frac{4\sqrt{3}}{3}$

25)

$\cos(\theta - \frac{\pi}{3})$ for $-\frac{\pi}{6}$ to $\frac{\pi}{3}$ decreases from 1 to 0

$\cos(\theta - \frac{\pi}{3})$ for $\frac{\pi}{3}$ to $\frac{5\pi}{6}$ increases from 0 to 1

At $\theta = \frac{\pi}{3}$ then $r = 3$

So the shortest distance to the line is $r = 3$ which is at angle $\theta = \frac{\pi}{3}$

A straight line where:

θ	0	$\frac{\pi}{6}$	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$
r	6	$2\sqrt{3}$	3	$2\sqrt{3}$	6

At $\theta = 0$ then $r = 6$

At $\theta = \frac{\pi}{2}$ then $r = 2\sqrt{3}$

26 a)

$r = \sinh\theta + \cosh\theta = \frac{1}{2}(e^{\theta} - e^{-\theta}) + \frac{1}{2}(e^{\theta} + e^{-\theta}) = e^{\theta}$

When $\theta = 0$, $r = 1$

When $\theta = \frac{\pi}{2}$, $r = e^{\pi/2}$

When $\theta = \pi$, $r = e^{\pi}$

26 b)

$r = \cosh\theta - \sinh\theta = \frac{1}{2}(e^{\theta} + e^{-\theta}) - \frac{1}{2}(e^{\theta} - e^{-\theta}) = e^{-\theta}$

When $\theta = 0$, $r = 1$

When $\theta = \frac{\pi}{2}$, $r = e^{-\pi/2}$

When $\theta = \pi$, $r = e^{-\pi}$

26 c)

$r = \sinh\theta - \cosh\theta = \frac{1}{2}(e^{\theta} - e^{-\theta}) - \frac{1}{2}(e^{\theta} + e^{-\theta}) = -e^{-\theta}$

When $\theta = 0$, $r = -1$

When $\theta = \frac{\pi}{2}$, $r = -e^{-\pi/2}$

When $\theta = \pi$, $r = -e^{-\pi}$

27 a)

When $-\frac{\pi}{2} < \theta \leq \frac{\pi}{2}$ then $0 < \cos\theta \leq 1$

So $0 < 3\cos\theta \leq 3$

Hence $0 < r \leq 3$

27 b)

When $0 < \theta \leq \frac{\pi}{2}$ then $0 < \sin\theta \leq 1$

So $0 < 2\sin\theta \leq 2$

Hence $0 < r \leq 2$

27 c)

When $0 < \theta \leq \frac{\pi}{2}$ then $0 \leq \cos\theta < 1$

So $3 \leq 3 + 2\cos\theta < 5$

Hence $3 \leq r < 5$

27 d)

When $-\pi < \theta \leq \pi$ then $-1 \leq \cos\theta \leq 1$

So $1 \leq 5 + 4\cos\theta \leq 9$

Hence $\frac{1}{3} \leq r \leq 3$

27 e)

When $0 < \theta \leq 2\pi$ then $-1 \leq \sin\theta \leq 1$

At $\theta = \frac{\pi}{2}$ then $\frac{3}{2-1} = 3$

At $\theta = \frac{3\pi}{2}$ then $\frac{3}{2+1} = 1$

The denominator is always positive

$1 \leq r \leq 3$

27 f)

When $0 \leq \theta \leq \frac{\pi}{2}$ then $0 \leq \cos\theta \leq 1$

At $\theta = 0$ then $\frac{3}{1-2\cos\theta} = \frac{3}{-1} = -3$

At $\theta = \frac{\pi}{2}$ then $\frac{3}{1} = 3$

$1 - 2\cos\theta = 0$ has a solution of $\theta = \frac{\pi}{3}$ for $0 \leq \theta \leq \frac{\pi}{2}$

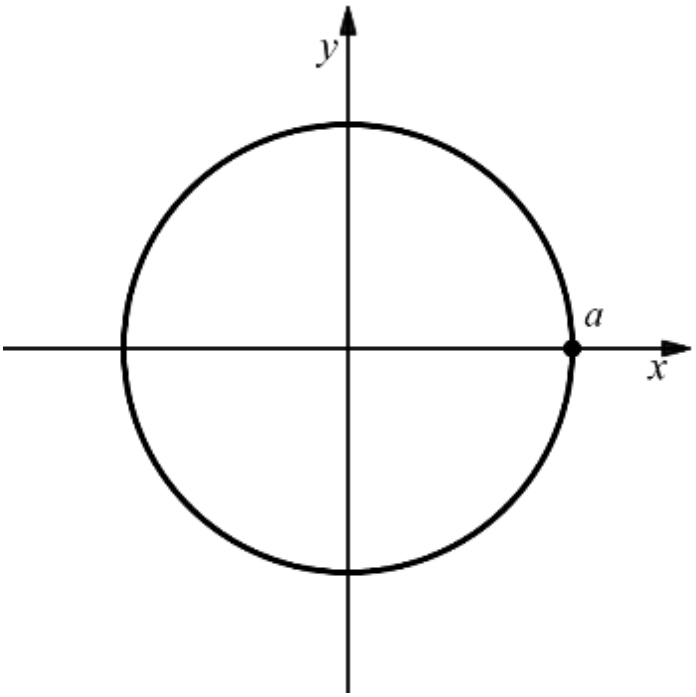
So $\frac{3}{1-2\cos\theta}$ has no solution for $\theta = \frac{\pi}{3}$

As $\theta \to \frac{\pi}{3}$, $\frac{3}{1-2\cos\theta} \to -\infty$

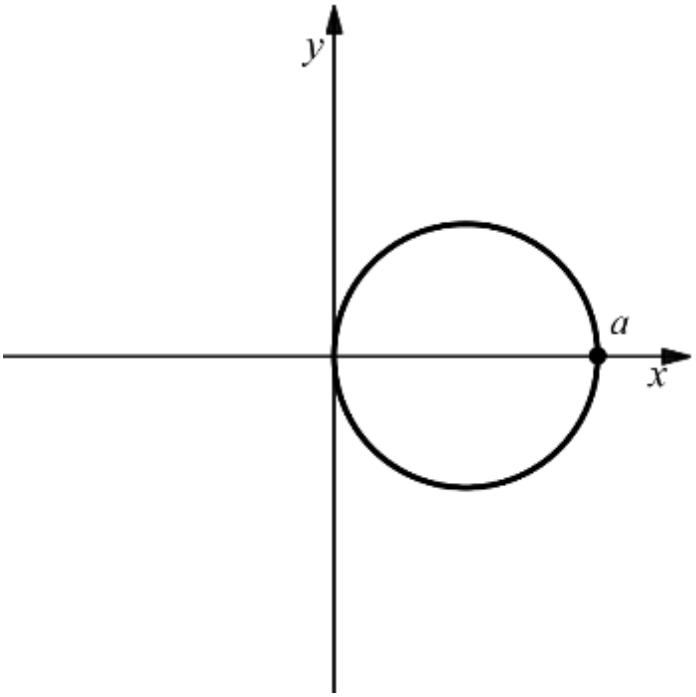
After $\frac{\pi}{3}$, θ is positive and decreasing to 3

Hence $r \leq -3$ or $r \geq 3$

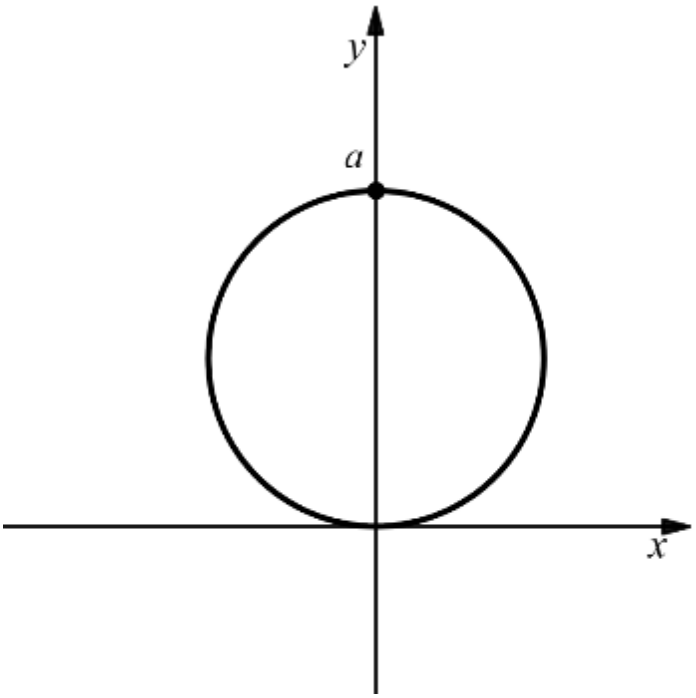
28 a)



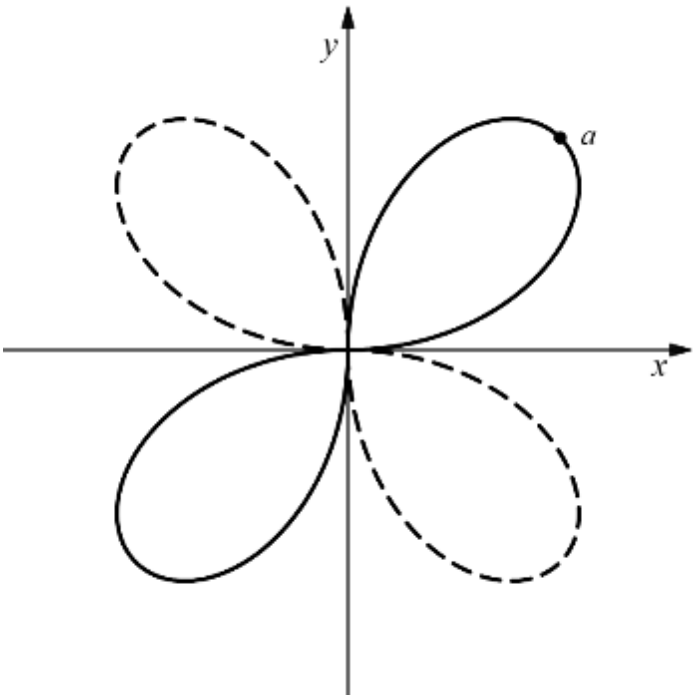
28 b)



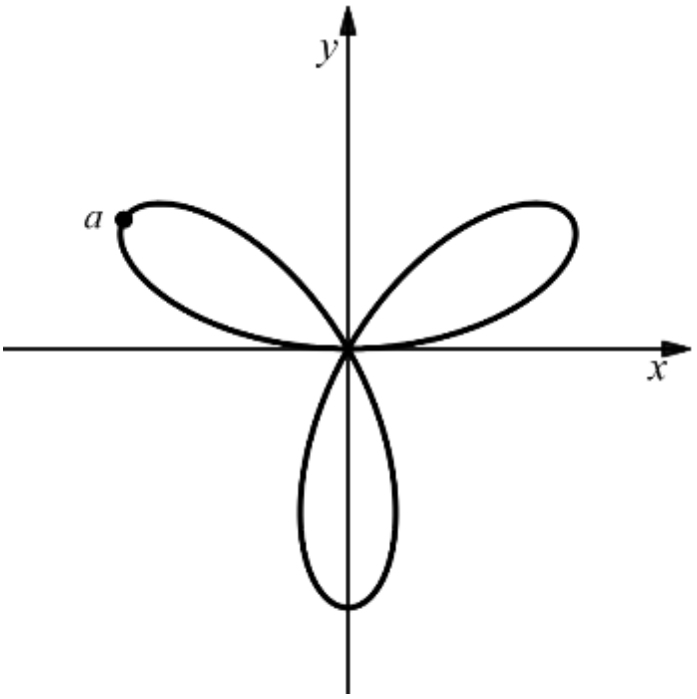
28 c)



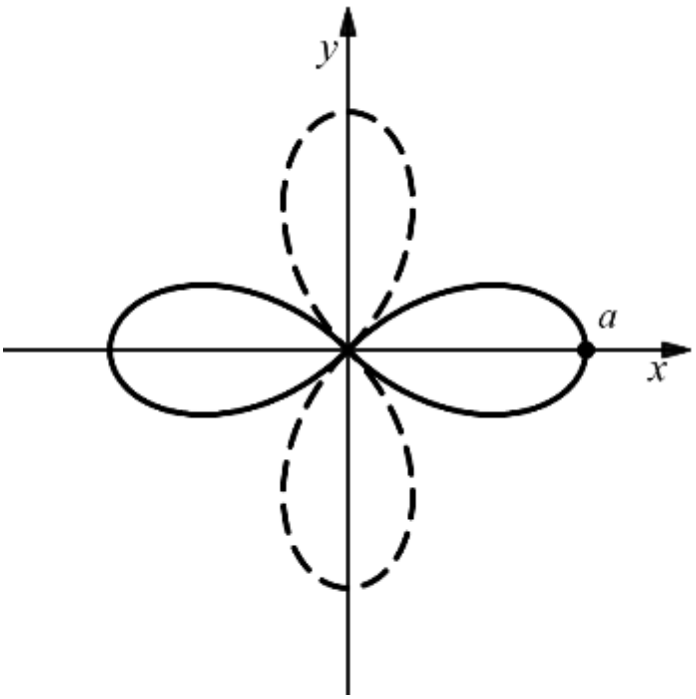
28 d)



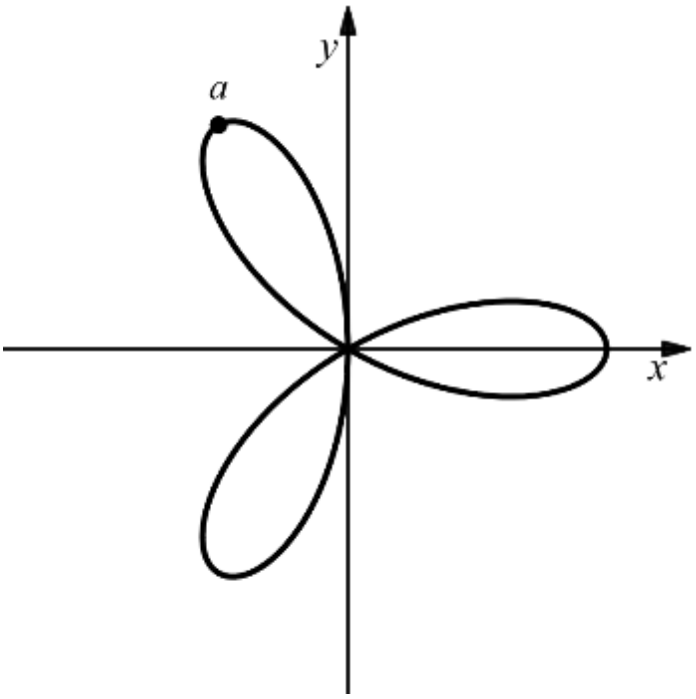
28 e)



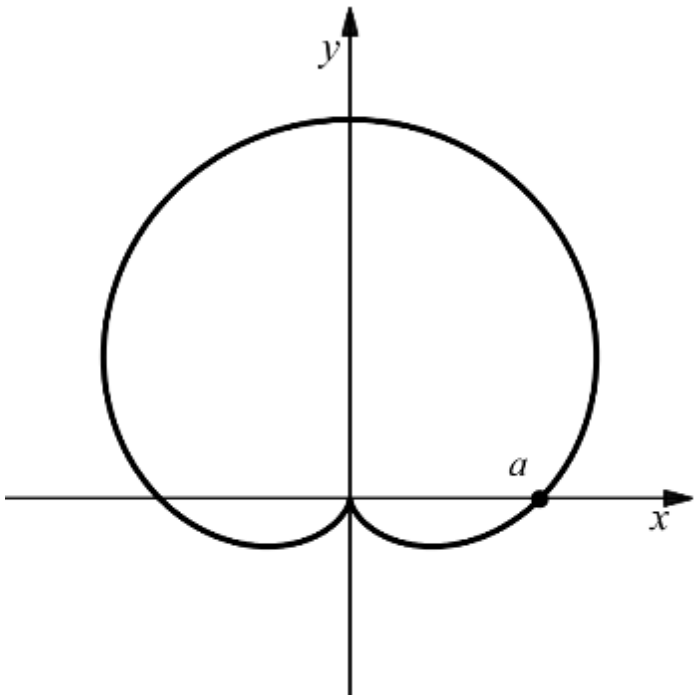
28 f)



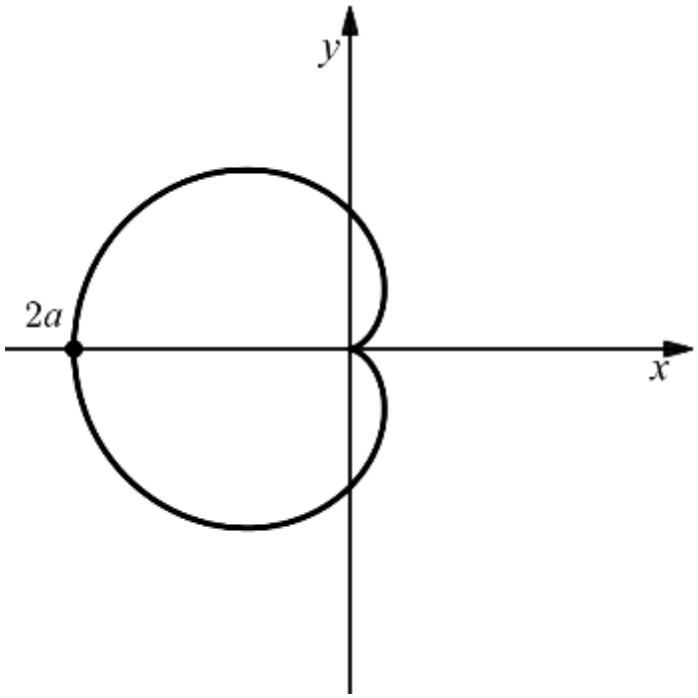
28 g)



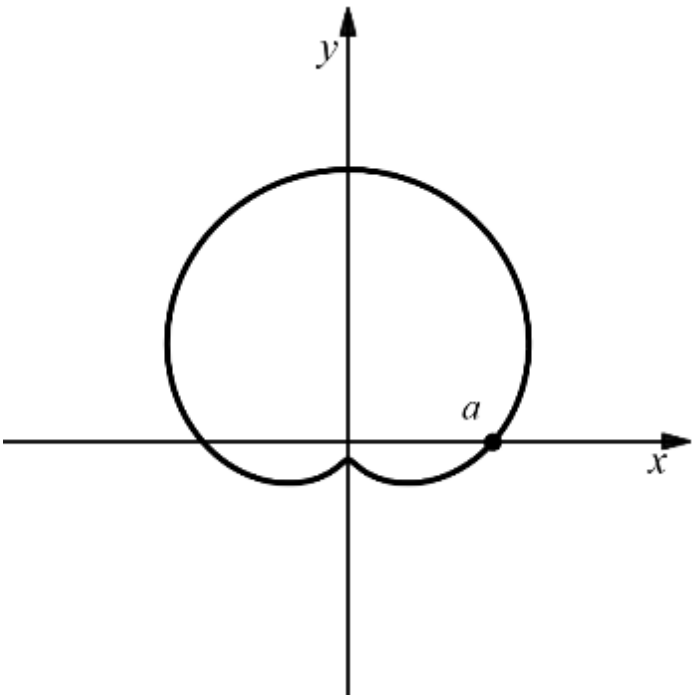
28 h)



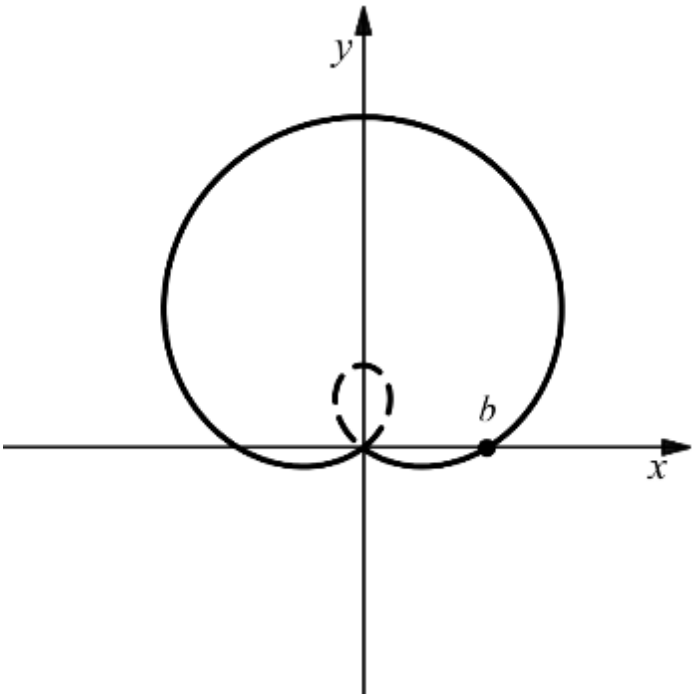
28 i)



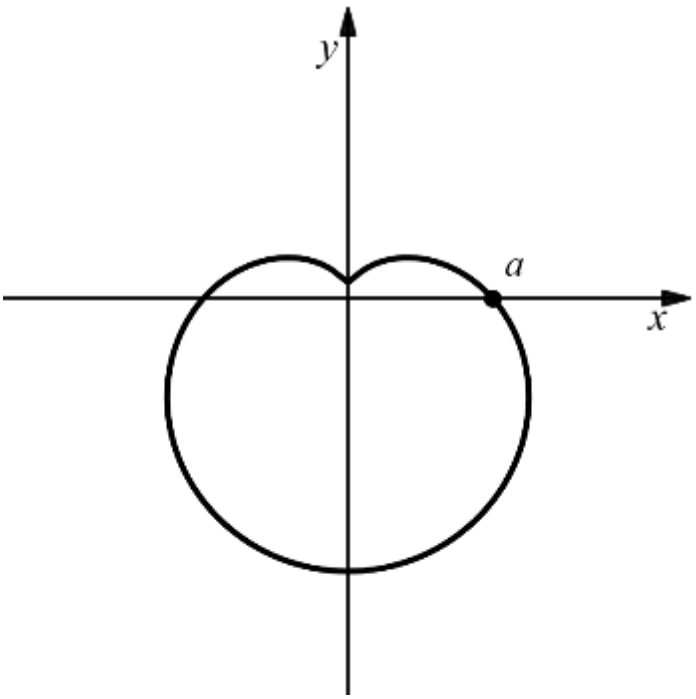
29 a)



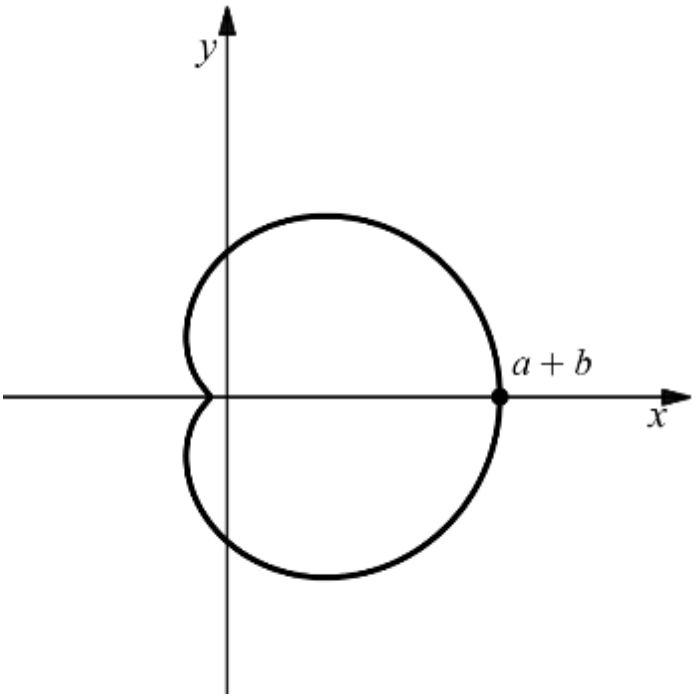
29 b)



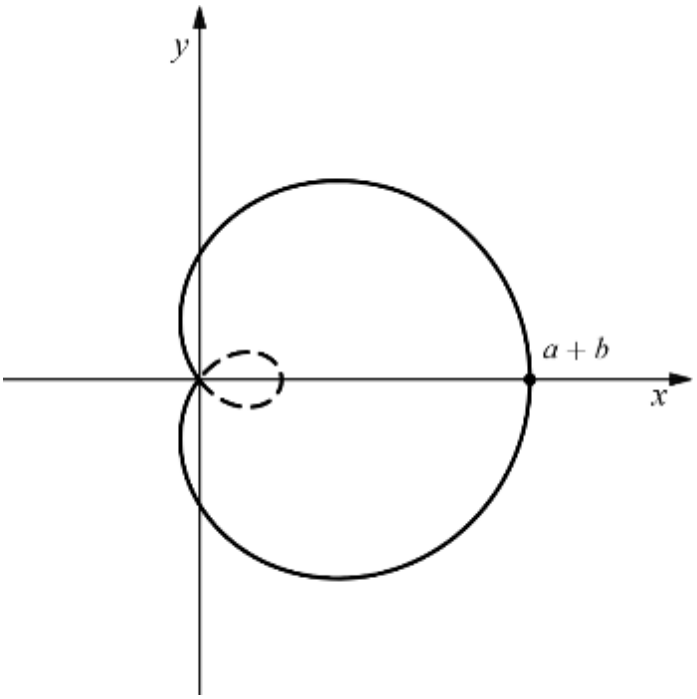
29 c)



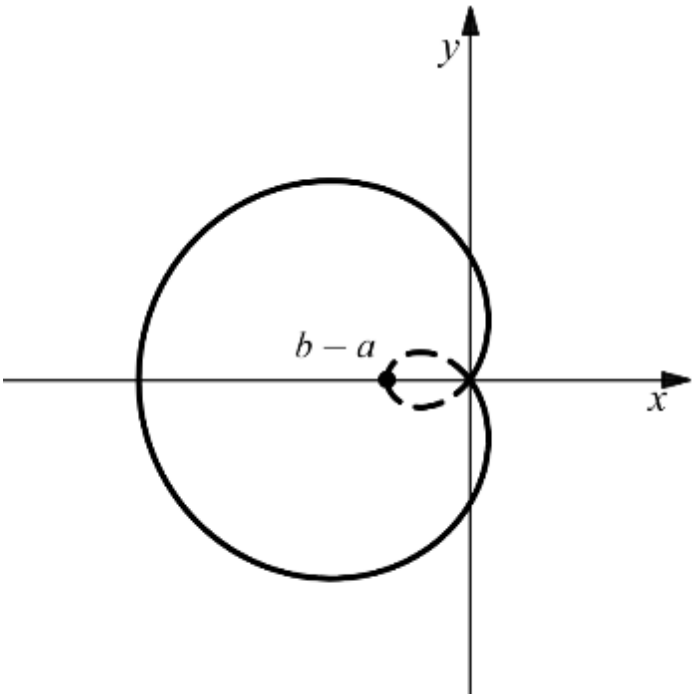
29 d)



29 e)



29 f)



30 a)

$2\theta = \theta + 2$

$\theta = 2$

So $(4, 2)$

30 b)

$3\theta = \theta + 1$

$\theta = \frac{1}{2}$

So $(\frac{3}{2}, \frac{1}{2})$

30 c)

$\theta^2 = \theta + 2$

$\theta^2 - \theta - 2 = 0$

$\theta = -1, \theta = 2$ so $\theta = 2$

So $(4, 2)$

31 a)

$r = 2(1 + \cos\frac{\pi}{2}) = 2$

$(2, \frac{\pi}{2})$

31 b)

$r = 2(1 + \cos 0) = 4$

$(4, 0)$

31 c)

$3 = 2(1 + \cos\theta)$

$3 = 2 + 2\cos\theta$

$\cos\theta = \frac{1}{2}$

$\theta = -\frac{\pi}{3}$ or $\theta = \frac{\pi}{3}$

$\therefore (3, -\frac{\pi}{3})$ and $(3, \frac{\pi}{3})$

32)

$2\cos^2\theta = 2 - 3\cos\theta$

$2\cos^2\theta + 3\cos\theta - 2 = 0$

$(2\cos\theta - 1)(\cos\theta + 2) = 0$

$\cos\theta = \frac{1}{2}$ or $\cos\theta = -2$

So $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$

$(\frac{1}{2}, \frac{\pi}{3})$ and $(\frac{1}{2}, \frac{5\pi}{3})$

33)

$3\sin^2\theta = 2 - 3\sin\theta$

$3\sin^2\theta + 3\sin\theta - 2 = 0$

$\sin\theta = \frac{-3\pm\sqrt{33}}{6}$

$\sin\theta = \frac{-3+\sqrt{33}}{6}$ or $\sin\theta = \frac{-3-\sqrt{33}}{6}$

$(0.628, 0.475)$ and $(0.628, 2.67)$

34)

$4\sin 2\theta = 4\sin\theta$

$4(2\sin\theta\cos\theta) = 4\sin\theta$

$2\sin\theta\cos\theta = \sin\theta$

$2\sin\theta\cos\theta - \sin\theta = 0$

$\sin\theta(2\cos\theta - 1) = 0$

$\sin\theta = 0$ or $\cos\theta = \frac{1}{2}$

So $\theta = \pi$, $\theta = 2\pi$ or $\theta = \frac{\pi}{3}$, $\theta = \frac{5\pi}{3}$

When $\theta = \pi$, $r = 0$ but $r > 0$

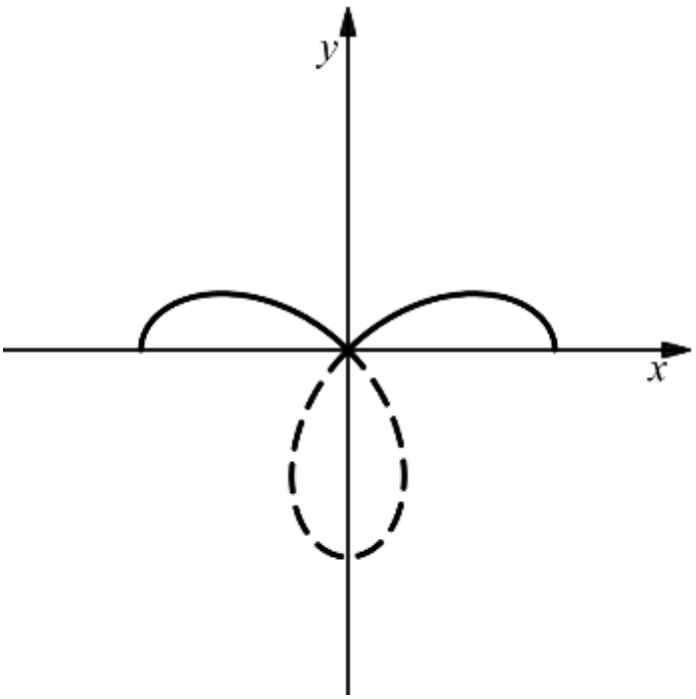
When $\theta = 2\pi$, $r = 0$ but $r > 0$

When $\theta = \frac{\pi}{3}$, $r = 2\sqrt{3}$

When $\theta = \frac{5\pi}{3}$, $r = -2\sqrt{3} \Rightarrow \theta = \frac{2\pi}{3}$, $r = 2\sqrt{3}$

$(2\sqrt{3}, \frac{\pi}{3})$ and $(2\sqrt{3}, \frac{2\pi}{3})$

35 a)



35 b)

$r = 3$

Expecting 2 solutions as $r = 6\cos 2\theta$ is defined for $0 \leq \theta \leq \pi$ and only 2 intersections occur when $r = 3$

The other 2 occur when $r = -3$

2 solutions from

$6\cos 2\theta = 3$

$\cos 2\theta = \frac{1}{2}$

$\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

So $(3, \frac{\pi}{6})$ and $(3, \frac{5\pi}{6})$

36 a)

When $\theta = \frac{\pi}{2}$ so $2a\sin^2(\frac{\pi}{2}) = 2a \times 1^2 = 2a$

When $\theta = -\frac{\pi}{2}$ so $2a\sin^2(-\frac{\pi}{2}) = 2a \times (-1)^2 = 2a$

So $P(2a, \frac{\pi}{2})$ and $Q(2a, -\frac{\pi}{2})$

36 b)

$2a\sin^2\theta = a(1 - \sin\theta)$

$2a\sin^2\theta + a\sin\theta - a = 0$

$2\sin^2\theta + \sin\theta - 1 = 0$

$(2\sin\theta - 1)(\sin\theta + 1) = 0$

$\sin\theta = \frac{1}{2}$ or $\sin\theta = -1$

$\theta = \frac{\pi}{6}$, $\theta = \frac{5\pi}{6}$ or $\theta = -\frac{\pi}{2}$

When $\theta = \frac{\pi}{6}$ so $2a\sin^2(\frac{\pi}{6}) = 2a(\frac{1}{2})^2 = \frac{a}{2}$

When $\theta = \frac{5\pi}{6}$ so $2a\sin^2(\frac{5\pi}{6}) = 2a(\frac{1}{2})^2 = \frac{a}{2}$

When $\theta = -\frac{\pi}{2}$ so $2a\sin^2(-\frac{\pi}{2}) = 2a(-1)^2 = 2a$

So $(\frac{a}{2}, \frac{\pi}{6})$, $(\frac{a}{2}, \frac{5\pi}{6})$ and $(2a, -\frac{\pi}{2})$